EVERY INFINITE SUBSET OF A NOETHERIAN SCHEME CONTAINS AN INFINITE SUBSET WITHOUT NONTRIVIAL SPECIALIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the separation of the points with no nontrivial specialization into T, the inductive definition of the strata with the dimension bound showing that only finitely many are nonempty, and the Noetherianity contradiction ruling out points outside the strata.

Editorial note. The label lemma-thin-infinite-sequence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source properties.tex –label lemma-thin-infinite-sequence –start 843 –end 893 –local-id 28032 returned RESULT=OK (L28032 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Finiteness of the dimension of the local rings of a Noetherian scheme. Algebra, the proposition that the dimension of the local ring of a Noetherian scheme at a point is finite. Supplies the finiteness of the maximum dimension d of the local rings at the points of the finite set T_0.
- Realization of the points specializing to a point inside its local ring. Schemes, the lemma that the points specializing to a given point are exactly the points of the spectrum of the local ring at that point. Supplies the identification of the points of a chain of specializations ending in T_0 as points of the spectrum of the local ring there, which bounds the length of the chain.
- Noetherianness of the underlying topological space. Same chapter, the lemma that a Noetherian scheme has a Noetherian underlying topological space. Supplies the impossibility of an infinite descending chain of specializations in the complement of the finite union.

Source reference. Original proposition 00KQ: sources/stacks-chow/tags.txt, source lines 760–760. Prerequisite pointer invoked at proof line(s) 868; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 01OY: sources/stacks-properties/pinned-original-source.tex, source lines 698–698. Prerequisite pointer invoked at proof line(s) 892; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01J7: sources/stacks-chow/tags.txt, source lines 2004–2004. Prerequisite pointer invoked at proof line(s) 876; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a Noetherian scheme. Let $T \subset S$ be an infinite subset. Then there exists an infinite subset $T' \subset T$ such that there are no nontrivial specializations among the points T'.*

Proof. Let $T_0 \subset T$ be the set of $t \in T$ which do not specialize to another point of T. If T_0 is infinite, then $T' = T_0$ works. Hence we may and do assume T_0 is finite. Inductively, for $i > 0$, consider the set $T_i \subset T$ of $t \in T$ such that

(1) $t \notin T_{i-1} \cup T_{i-2} \cup \ldots \cup T_0$,
(2) there exist a nontrivial specialization $t \rightsquigarrow t'$ with $t' \in T_{i-1}$, and
(3) for any nontrivial specialization $t \rightsquigarrow t'$ with $t' \in T$ we have $t' \in T_{i-1} \cup T_{i-2} \cup \ldots \cup T_0$.

Again, if T_i is infinite, then $T' = T_i$ works. Let d be the maximum of the dimensions of the local rings $\mathcal{O}_{S,t}$ for $t \in T_0$; then d is an integer because T_0 is finite and the dimensions of the local rings are finite by Algebra, Proposition 00KQ. Then $T_i = \emptyset$ for $i > d$. Namely, if $t \in T_i$ then we can find a sequence of nontrivial specializations $t = t_i \rightsquigarrow t_{i-1} \rightsquigarrow \ldots \rightsquigarrow t_0$ with $t_0 \in T_0$. As the points $t = t_i, t_{i-1}, \ldots, t_0$ are in $\mathrm{Spec}(\mathcal{O}_{S,t_0})$ (Schemes, Lemma 01J7), we see that $i \leq d$. Thus $\bigcup T_i = T_d \cup \ldots \cup T_0$ is a finite subset of T.

Suppose $t \in T$ is not in $\bigcup T_i$. Then there must be a specialization $t \rightsquigarrow t'$ with $t' \in T$ and $t' \notin \bigcup T_i$. (Namely, if every specialization of t is in the finite set $T_d \cup \ldots \cup T_0$, then there is a maximum i such that there is some specialization $t \rightsquigarrow t'$ with $t' \in T_i$ and then $t \in T_{i+1}$ by construction.) Hence we get an infinite sequence

$$t \rightsquigarrow t' \rightsquigarrow t'' \rightsquigarrow \ldots$$

of nontrivial specializations between points of $T \setminus \bigcup T_i$. This is impossible because the underlying topological space of S is Noetherian by Lemma 01OY. $\square$